Advancing Pharmaceutical Distribution Through Data-Driven Inventory Optimization, Demand Forecasting, and Retailer Performance Analytics

A Midterm Report for the BDM Capstone Project

Submitted by

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Executive Summary:-

Sevam Pharma LLP is a pharmaceutical distribution firm based in Sangli, Maharashtra. The company operates in the B2B segment and supplies medicines to retail medical stores in the surrounding region. Most of the medicines distributed by the firm are sourced from Sun Pharma and include products used for neurological and psychiatric treatments. The organization manages procurement, warehouse inventory, and distribution of medicines to multiple retailers across its operating region.

As the business expanded and the number of products and retailers increased, the company began facing several operational challenges. Some slow-moving medicines remain in stock for long periods, increasing the risk of expiry and financial loss, while fast-moving medicines may go out of stock and lead to missed sales opportunities. In addition, inefficient stock allocation across retailers sometimes prevents the company from fully capturing market demand. Even when demand exists, some retailers may not receive sufficient stock while others hold excess inventory. Another issue is inconsistencies or anomalies in stock and sales records, which makes it difficult to clearly track product movement.

To better understand these issues, the available stock and sales data will be collected and prepared for analysis. The preprocessing stage will involve organizing the dataset, cleaning inconsistencies in the records, defining metadata for the variables, and calculating descriptive statistics to understand the overall structure of the data. After preparing the dataset, exploratory analysis will be carried out to examine product-wise and retailer-wise sales patterns and inventory movement. Techniques such as ABC analysis and FSN analysis may be used to classify medicines based on value and movement, while trend analysis can help observe demand patterns over time. These analyses can help identify fast-moving and slow-moving products, highlight possible inventory imbalances across retailers, detect unusual variations in stock or sales data, and support better decisions related to procurement, stock allocation, and overall inventory management.

Proof of Originality:-

- **Organization Address:**
  Kubera Chambers, Amrai Rd, near Officers Club, Vakhar Bhag, Sangli, Maharashtra 416416
- **Recorded interaction with the owner (Link):**
  https://drive.google.com/drive/folders/1xNe_Zxwu7vwWJsFfQFaIfLIiEs1xsBgm?usp=sharing
- **Images of the organization:**



- **Authorization Letter:**

SEVAM PHARMA LLP

Pharmaceutical Distributors
Kubera chambers, Vakhar Bhag,
Sangli, Maharashtra 416416

Feb 20, 2026

To:
Abhay Sharma
23F3001400
IIT Madras

I am writing on behalf of SEVAM PHARMA LLP to formally authorize your use of our business data for your project, "Advancing Pharmaceutical Distribution Through Data-Driven Inventory Optimization, Demand Forecasting, and Retailer Performance Analytics".

This authorization is valid from March 2026 to December 2026 and is limited to the scope of your project. The organization will be providing comprehensive data spanning from Feb 2025 to Jan 2026. Please always ensure the confidentiality and security of the data and refrain from sharing it with third parties without our explicit written consent.

Upon completion of your project, kindly delete or return all data provided by SEVAM PHARMA LLP as per our instructions. We look forward to the insights your project will generate and remain available for any further assistance you may need.

Regards,

K.Ajit
PROPRIETOR
SEVAM PHARMA LLP
LLPIN: ABA-2176

Ajit Anna Khavate
Designated Partner

Metadata:-

Metadata refers to a brief description of the data, often called "data about data," which helps in understanding the background and structure of the dataset. In this project, metadata includes two aspects: metadata of the business organization and metadata of the dataset used for analysis. The business metadata provides information about the organization such as establishment details, ownership, type of business, and operational activities [Table-1]. The dataset metadata describes the structure of the collected data, including variables related to sales, purchases, stock levels, revenue, and other operational records. Together, this information provides the necessary context to understand the source, structure, and purpose of the data before performing further analysis.

- ***Metadata of business:-***

S. No.	Key Metric	Value
1	Organization Name	Sevam Pharma LLP
2	Location	Sangli, Maharashtra
3	Owners / Partners	Ajit Anna Khavate, Shridhar Babasaheb Dhamnikar, Avinash Krishnaji Pore, Sandeep Ashokrao Patil, and Sachin Dhanchandra Sakale
4	Year of Establishment	2022
5	LLPIN	ABA-2176
6	Organization Type	B2B Pharmaceutical Distributor
7	Number of Employees	12
8	Business Activity	Procurement, storage, and distribution of medicines to retail medical stores
9	Main Supplier	Sun Pharma
10	Area of Operation	Sangli and nearby regions
11	Key Issues Addressed	Stock imbalance, expiry risk of medicines, stock shortages of fast-moving medicines, inefficient stock allocation across retailers

Table 1: Representation of business metadata

- ***Metadata of data used in analytics:-***

The organization has provided data of 1 year from February 2025 to January 2026.
There are 3 datasets for each month, which are as below:

1. Product-wise Sales Data
2. Product-wise Purchase Data
3. Monthly Stock Statement

For the purpose of analysis, the monthly datasets were combined and organized into consolidated datasets. Product-wise sales data as well as purchase data from all months were merged to form annual sales-purchase dataset, and stock records from each month were combined into a unified stock dataset.

The key features of each dataset type are mentioned and explained below:

i. ***Product-wise Sales Data:***

   a. ***Customer, Product, Batch, Expiry, invDt, invNo*:** Represent the retailer name, product name, batch number, expiry date, invoice date, and invoice number respectively.

   b. ***Qty, Free, MRP, Value*:** Represent the quantity purchased, free quantity provided under schemes or promotions, maximum retail price of the product, and total sales value respectively.

   c. ***pDisc, bDisc, sRate, trRate*:** Represent the manufacturer's discount to the company, company's discount to retailers, sale rate to retailers, and purchase rate from manufacturers respectively.

ii. ***Product-wise Purchase Data*:**

   a. ***Plant, Plant Desc, Material Code ,Material Code Desc, Batch, Expiry*:** Represent the plant code, plant description, product code, product name, batch code and expiry date respectively.

   b. ***Unit, Timestamp, Type*:** Represent the unit of measurement of the product (e.g., ml for syrups and nos for tablets or capsules), the date and time of the transaction, and the type of transaction (invoice or return), respectively.

   c. ***Billed Quantity, Net Amount, Purchase Rate*:** Represent the quantity of product purchased, the total purchase amount and price per quantity respectively. Both quantity and amount can take negative values as well if the type of transaction is "return".

iii. ***Monthly Stock Statement*:**

   a. ***Product Name, Pack, Rate*:** Represent the name of the product, packaging size of the product, and the rate per pack respectively.

   b. ***Previous Sale, Opening, Receipt*:** Represent the total sales quantity in the previous period, opening stock at the beginning of the month, and quantity of stock received during the month respectively.

   c. ***Sales, Free, Sales RTN, Purchase RTN, Others*:** Represent the quantity sold, quantity provided for free under promotional schemes, quantity returned by retailers, quantity returned to suppliers, and other stock adjustments respectively.

   d. ***Sale Value, Closing, Cl Value, expQty, nonMoveQty*:** Represent the total sales value of the product, closing stock quantity at the end of the month, total value of the closing stock, expired stock quantity, and non-moving stock quantity respectively.

Descriptive Statistics:-

Descriptive Statistics involves summarizing the dataset and explaining the characteristics of the variables involved. In this analysis, descriptive statistics provide insights such as the average sales per month, average revenue generated, variation in profit over time, and the minimum and maximum values of operational metrics such as purchases and purchase value.

One of the most commonly used descriptive statistical methods is the ***Five-Number Summary***, which consists of the ***minimum, first quartile (Q1), median, third quartile(Q3) and maximum.*** To enhance the analysis, additional statistical metrics are included: ***Mean*** (average value representing central tendency), ***Variance and Standard Deviation*** (measuring the dispersion or variability of the data), and ***Skewness*** (indicating whether the distribution is symmetric, left-skewed, or right-skewed).

Metric	No_of_Sales	Revenue	Profit	No_of_Purchases	Purchase_value
Min	5548	1543797.46	310579.59	115	1107218.94
1st Quartile	6154.5	1755533.3	353238.13	127.5	1295329.74
Median	6937	1942001.38	384051.71	135.5	1394946.48
3rd Quartile	7437	2175332.4	439176.27	138.25	1583109.83
Max	8099	2259878.23	452234.3	159	2209783.68
Mean	6885.25	1933832.77	386358.73	134.17	1481563.29
Variance	673083.3	56032795601	2519581543	172.88	1.05391E+11
Std.Dev	820.42	236712.47	50195.43	13.15	324639.4
Skew	0.04	-0.16	-0.11	0.25	1.11

Table 2: Representation of Descriptive Statistics

In addition to descriptive statistics, understanding the relationship between variables helps identify operational patterns. This relationship is measured using the **Correlation Coefficient**, which ranges from **-1 to 1**, where +1 indicates a strong positive relationship, -1 indicates a strong negative relationship, and 0 indicates no relationship between the variables.

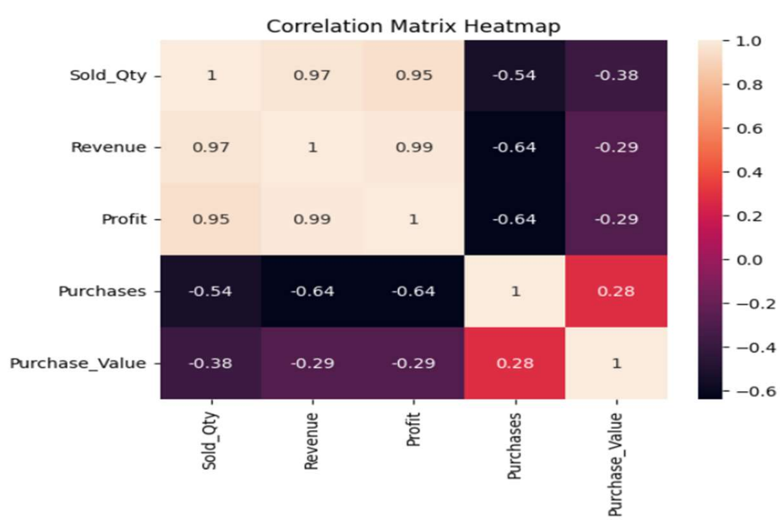

Figure 1: Correlation Matrix Heatmap

The descriptive statistics shows that average monthly sales are about **6,885 units**, generating **₹19.33 lakh revenue** and **₹3.86 lakh profit**, with moderate variation across months. Purchasing activity averages around **134 transactions per month** with a purchase value of about **₹14.82 lakh**, showing some fluctuations [Table 2]. Correlation analysis indicates that higher sales volume directly increases revenue and profit, while purchase value rises with the number of purchase transactions, reflecting the relationship between sales and procurement activities [Figure 1].

Detailed Explanation of Analysis Process:-

The steps explain how the data was cleaned and analyzed, and the importance of analysis that was performed.

- ***Data Cleaning:***

  This part includes checking the format in which the data is being provide and identifying the appropriate columns and checking for unwanted columns, checking for missing values. The data contain some duplicated columns and some columns which are not necessary for our analysis. Some such columns are:

  - In the product sales data, columns ***pDisc*** and ***bDisc*** are having value zero for all the sales, so they don't contribute anything.
  - In the product purchase data, columns ***Plant*** and ***Plant Desc*** are having values ***"7157"*** and ***"7157-SPDL CFA-BHIWANDI"*** for all the rows since all the medicines are coming from the same manufacturer, ***SUN Pharma***.
  - Fields like ***batch no.*** and ***expiry date*** are present both in sales as well as purchase data, thus are redundant.

  In the product purchase data, the value of ***Material Code*** is missing for some of the medicines, since it is a unique identifier for the medicine we are imputing it's value with zero to maintain consistency.

- ***Data Analysis:***

  After completing the data cleaning and preprocessing stage, the monthly datasets were combined to create a consolidated ***annual dataset***. This allows us to analyze the pharma's performance across the entire period instead of looking at each month separately. The analysis was carried out using ***Python***, mainly with libraries such as ***Pandas*** for handling and manipulating data, and ***Matplotlib*** and ***Seaborn*** for creating visualizations.

One of the analyses performed is **ABC Analysis**, which groups medicines based on how much they contribute to the total sales value. In this method, medicines are divided into three categories: **A, B, and C**. Category **A** includes medicines that contribute the highest share of revenue, **B** includes medicines with moderate contribution, and **C** includes medicines that contribute the least. This method is more appropriate than treating all products equally because it prioritizes items based on their financial importance. This analysis helps warehouse managers focus more attention on high-value medicines so that they are always available in stock and properly managed.

Another important method used is **FSN Analysis (Fast, Slow, Non-moving)**. This analysis classifies medicines based on how frequently they are sold. **Fast-moving medicines** are sold regularly and need frequent restocking, **slow-moving medicines** are sold occasionally, and **non-moving medicines** are rarely sold. This method is particularly appropriate for pharmaceutical data because it directly considers product movement and expiry risk, which many general inventory methods do not address. Identifying slow or non-moving medicines helps reduce unnecessary stock and prevents wastage.

I also conducted a **Sales Trend Analysis** to understand how sales change over time. By examining sales values and quantities across the dataset, patterns such as periods of higher demand or changes in overall sales performance were observed. This method is more suitable than static analysis because it captures temporal changes and demand fluctuations. These insights help in planning inventory purchases more effectively and preparing for changes in demand.

Another analysis performed is **Retailer-wise Sales Analysis**, which looks at how much each retailer contributes to the overall sales. This helps identify key retailers who generate a large portion of the revenue. Compared to product-only analysis, this method provides insight into customer contribution and demand distribution across retailers. Understanding which retailers contribute the most can help the pharmacy build stronger relationships with important customers and plan better distribution or sales strategies.

Results and Findings:-

Visualizations helps us understand better. Charts like Bar Chart, Line Chart, Bubble Chart etc are used.

- **Link to dataset and analysis:** https://drive.google.com/drive/folders/1nWzxI0YegzP94zi_RgduMiYuFr8ooxCm?usp=sharing

Note: Due of space constraint, some of the important visualisations are shown here whereas the remaining are provided in the drive along with the cummulative dataset and python code.

ABC and FSN Category Distributions:

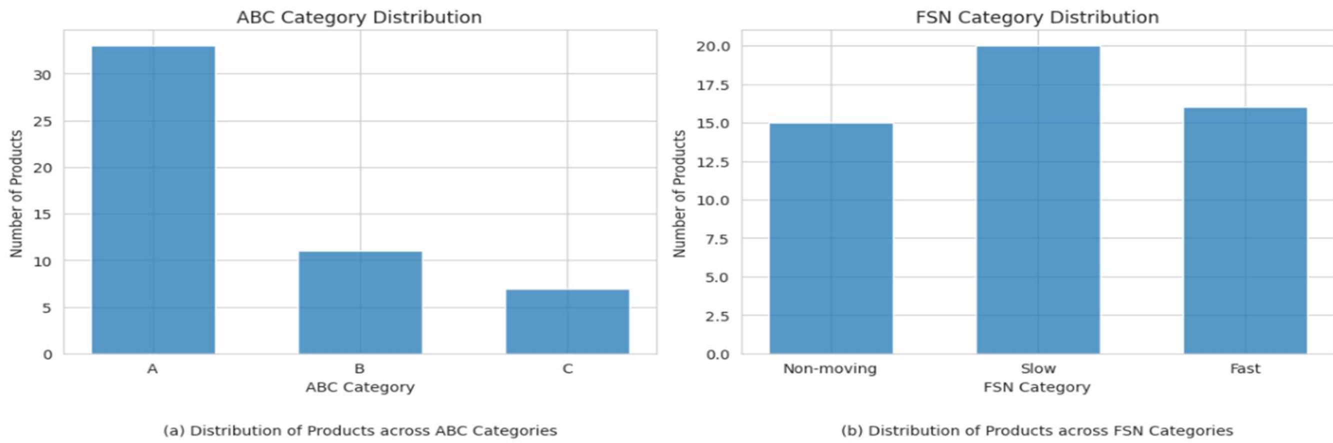

Figure 2: Bar Graphs of ABC and FSN Analysis

ABC analysis shows that most products fall into Category A, indicating that a smaller set of medicines contributes significantly to revenue [Figure 2(a)]. FSN analysis reveals many slow or non-moving items, which occurs because distributors must maintain a wide range of medicines even if some are rarely demanded by retailers [Figure 2(b)].

Bubble Chart of Revenue vs Month:

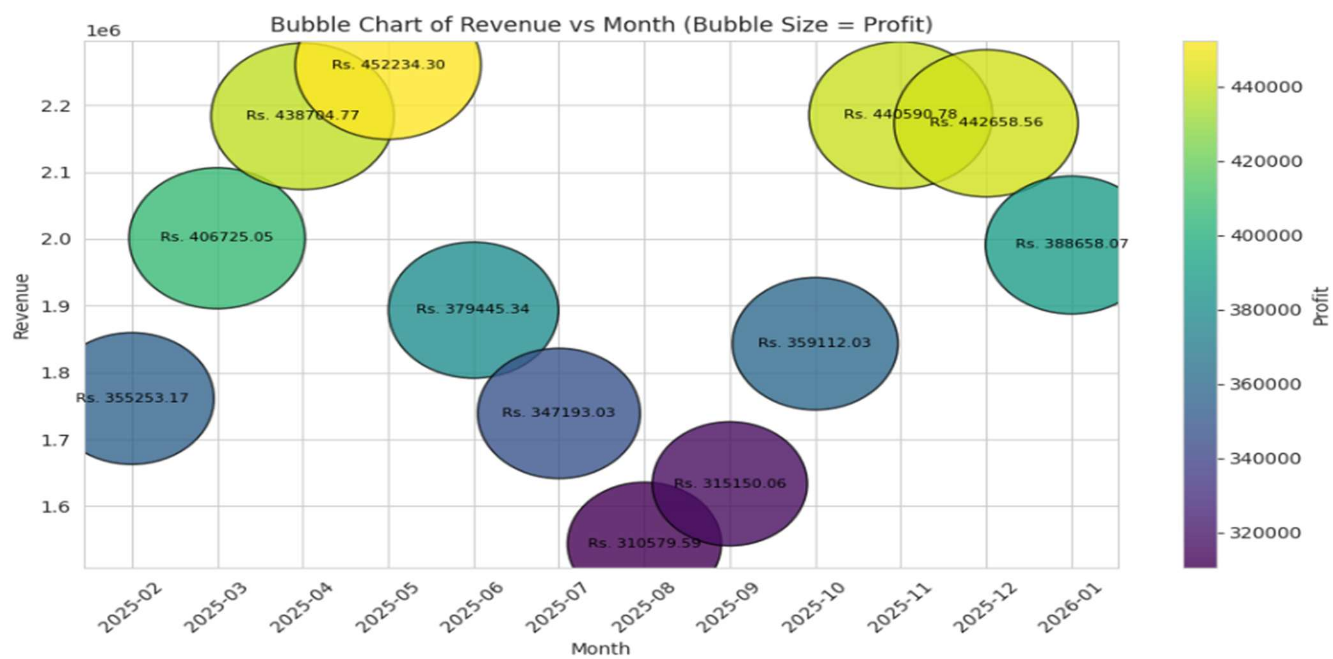

Figure 3: Bubble Chart of Revenue vs Month

Months with higher revenue also show larger bubbles, indicating higher profits because profit is directly derived from sales volume. Peaks around April–May and November–December likely reflect higher medicine demand from retailers, while the dip around August–September may be due to lower seasonal demand or reduced retailer stocking [Figure 3].

Profit and Revenue Trend:

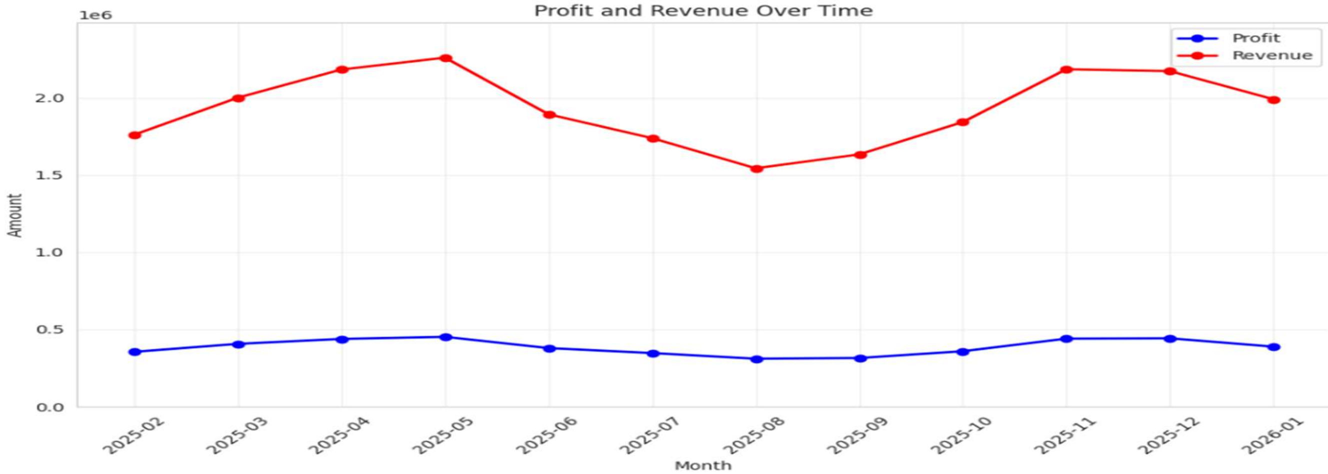

Figure 4: Profit and Revenue Over The Months

Revenue and profit follow a similar pattern across months, indicating a stable profit margin. As revenue increases, profit rises proportionally because each sale contributes consistent margins. The mid-year decline suggests reduced retailer purchases, while the later rise reflects increased demand and higher medicine sales toward the end of the period [Figure 4].

Top 10 Retailers by Revenue:

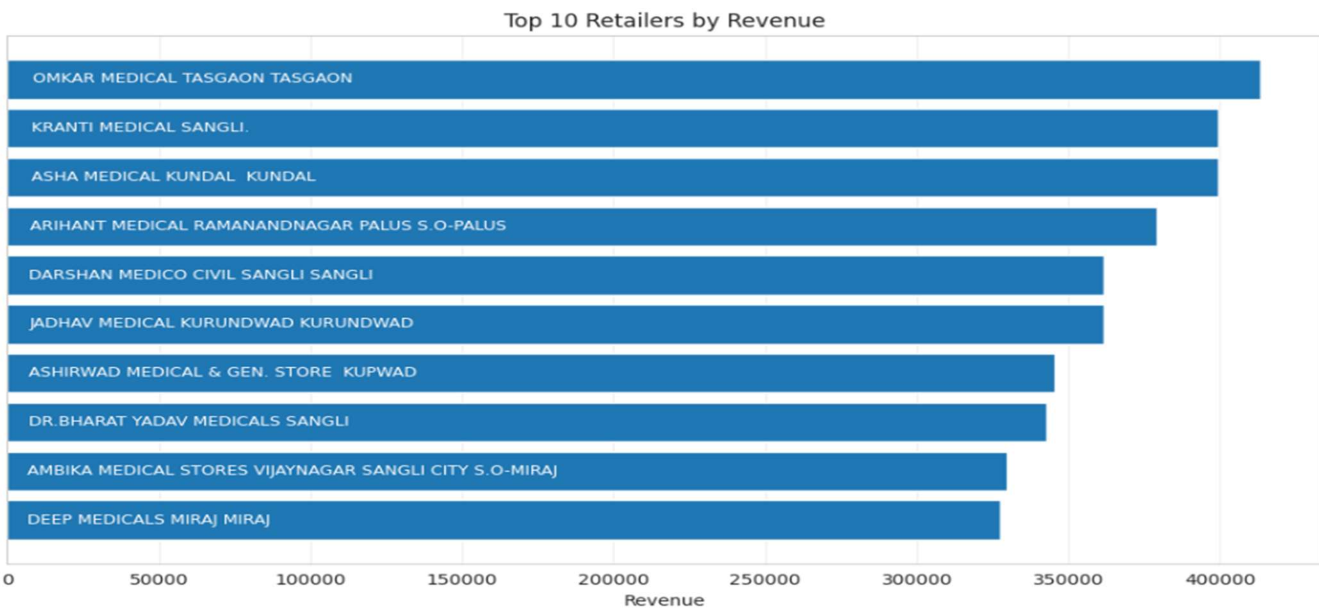

Figure 5: Bar Plot of Retailer vs Revenue (Top 10)

A small group of retailers contributes the largest share of revenue, showing a concentrated sales distribution. This pattern occurs because larger or more active medical stores place higher order volumes and purchase frequently, making them key customers who significantly influence overall sales performance [Figure 5].